

## No. of Functional Features Included in the Solution

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 01 July 2026                               |
| <b>Team ID</b>       | XXXXXX                                     |
| <b>Project Name</b>  | Voice-Based Concept Understanding Analyser |
| <b>Maximum Marks</b> | 5 Marks                                    |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name                  | Feature Description                                                   | Module / Component              | Status (Done / In Progress / Pending) | Marks Contribution |
|------|-------------------------------|-----------------------------------------------------------------------|---------------------------------|---------------------------------------|--------------------|
| 1    | Audio Upload & Recording      | Upload or record voice responses directly on the UI dashboard.        | Frontend / Audio Input          | Done                                  | Medium             |
| 2    | Speech Transcription Engine   | Automatically converts raw spoken audio files into plain text         | Backend / Speech Recognition    | Done                                  | High               |
| 3    | AI Concept Evaluation         | Scores the response text out of 100 based on core accuracy.           | Backend / Gemini AI Integration | Done                                  | High               |
| 4    | Automated Feedback Generation | Provides detailed, qualitative feedback points on strengths and gaps. | Backend / Generative AI         | Done                                  | High               |
| 5    | Fluency & Speech Metrics      | Computes Words Per Minute (WPM), word count, and filler words.        | Backend / Core Logic Engine     | Done                                  | Medium             |
| 6    | Interactive Waveform Charting | Plots and displays an interactive audio waveform graph.               | Backend / Data Visualization    | Done                                  | Low                |
| 7    | Evaluation History Logs       | Securely saves and fetches past user test scores from database.       | Database / SQLite3              | Done                                  | Medium             |
| 8    | Automated PDF Reporting       | Compiles metrics and chart into a downloadable PDF summary.           | Backend / ReportLab PDF Engine  | Done                                  | High               |

### Feature Summary:

| Metric                 | Count / Value |
|------------------------|---------------|
| Total Features Planned | 8             |

|                                    |   |
|------------------------------------|---|
| Total Features Implemented         | 8 |
| Core / Must-Have Features          | 8 |
| Additional / Nice-to-Have Features | 0 |
| Features Tested & Verified         | 8 |

**Feature Category Breakdown:**

| S.No | Category                  | Features in Category | Example Features                                                                              |
|------|---------------------------|----------------------|-----------------------------------------------------------------------------------------------|
| 1    | User Interface (UI)       | 3                    | Audio Recording Dashboard, Past Records History Table, Interactive Waveform Display.          |
| 2    | Backend / Logic           | 3                    | Speech-to-Text Conversion, WPM & Filler Words Calculation, ReportLab PDF Document Generation. |
| 3    | Database / Storage        | 1                    | SQLite3 Database Integration for storing audio evaluation history log records.                |
| 4    | API / Integration         | 1                    | Google GenAI SDK Integration for Gemini AI connection and score generation.                   |
| 5    | Security / Authentication | 1                    | Basic user/student identifier management for local session tracking.                          |